

What Should SRE Watch?

Calibration-Gated Decisions and the Missing Golden Signals for GPU Clusters

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scan for live demo

Queue wait is the user-visible latency on a shared GPU cluster. GPU utilization does not see it. We measure Queue-of-Work Reliability (QWR) — a calibrated $P(\text{wait} > \text{SLO})$ — and gate actions on its ECE. The live production model is at ECE 0.068: *warn*, not admit.

THE FOUR TRAVEL POORLY

Classical golden signals

Google SRE's four — latency, traffic, errors, saturation — assume a state-less request/response service. On shared GPU clusters each category needs a different operational definition.

Latency has token, batch, epoch, *and queue* meanings. Submit-to-start often dominates. HTTP p99 never sees it.

Traffic as QPS is undefined for training. Tokens/s and active jobs are different loads.

Errors as HTTP 5xx miss silent quality failure (200 + a bad output) and OOM / throttle restarts.

Saturation as `nvidia-smi` utilization is time-busy, not useful compute. Memory-bound kernels, thermal trips, and NCCL stragglers can all look fully utilized.

These are mechanistic failure modes, not r -values from this trace. The Slurm DCGM slice is idle T4 queue-wait (utilization = 0), so a util-goodput correlation is undefined.

A PAGING-ROTATION MINIMUM

Five candidate signals

QWR latency / queue At submit: $\hat{p} = P(\text{wait} > \text{SLO})$. Rolling ECE of $\hat{p}$. Advisory ≤ 0.05 ; gate ≤ 0.03 . Measured — held-out Slurm + live VGAC
GCE saturation / compute Util dispersion, thermal spread, power-brake rate. Utilization is one feature, not the signal. Defined — needs non-idle DCGM
HPH saturation / HBM Headroom remaining, plus its derivative, to catch OOM cliffs before they page. Defined — same idle-trace limit
SAI latency / collectives P_{90}/P_{50} step time across workers. “Goodput = 1/SAI” only if steps are fully synchronous. Candidate — no per-worker step times
OQC errors / quality ECE of <i>model outputs</i> vs a shadow label stream. Not the queue-delay classifier. Candidate — not measured here

THE TITLE CLAIM

The gate currently warns

Actions are not “the model said 70%.” They are gated on whether that 70% is calibrated.

T1 Observe always on	T2 Suggest held-out Slurm	T3 Warn live ECE 0.068	T4 Gate ECE ≤ 0.03
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Live VGAC

v4.0-richfeatures-lr

ECE **0.068** AUROC 0.766 $n = 5,184$

Fails 0.05 advisory and 0.03 gate.

Warn tier — not T4 Gate.

Held-out Slurm

public slurm_real_benchmark.json

444 train / 111 test of 555 jobs

GB AUROC **0.863** (CI 0.71–0.96)

ECE **0.056** tier **suggest**

~8 positives in test 0 true alerts at $p \geq 0.7$

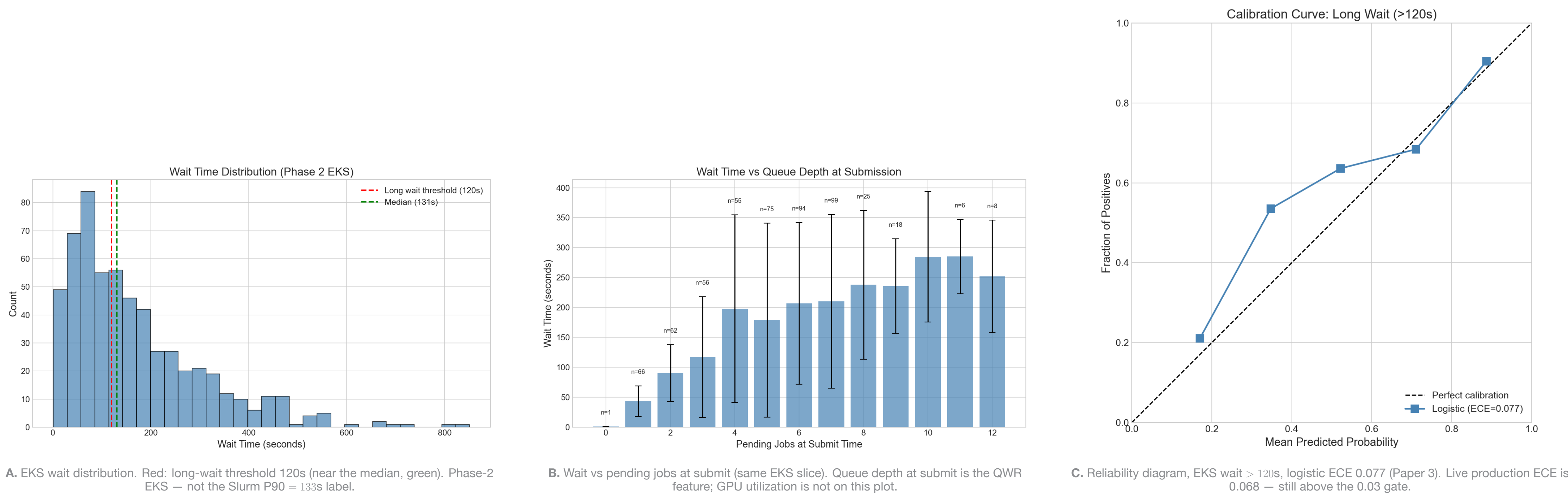
Labels are not the same

- EKS long wait: wait > 120s (near the median).
- Slurm long wait: wait > P90 = 133s (~10% base rate). Do not pool AUROC or ECE across those labels.

Paper 3 abstract (650 EKS jobs, ECE 0.077) is figure C — not the live 0.068 card.

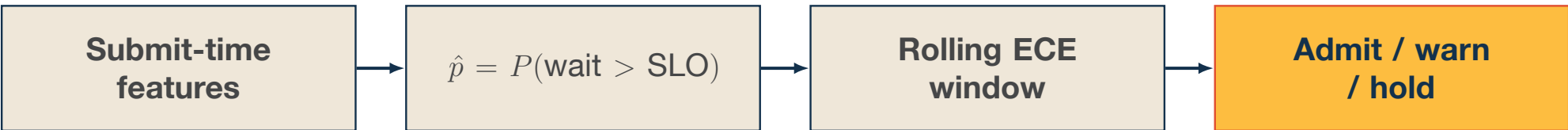
WHAT WE CAN PLOT

Queue wait is the measured story



HOW AN SRE WOULD USE THIS

From $\hat{p}$ to a decision



VGAC at demo.vgac.cloud is that loop on a t4g.small Graviton host. Predictions = QWR. Calibration = the gate (currently warn). GPUs = GCE/HPH candidates. HPC = SAI candidate.

DO NOT PRINT

Limits

- Not a five-signal validation on four environments.
- Do not cite local-only AUROC 0.985 / ECE 0.006.
- Do not print the six-panel SLI dashboard (wrong EKS tier, empty panel, degenerate ECE).
- Drift PSI ≈ 12.4 is an empty-bin artifact.
- 15–25% goodput loss to stragglers is literature, not this trace.